

Problem statement of your project/ Title of your Project

Project Domain: Internet of Things

SYBTech Project Synopsis

Bachelor of Technology

IN

Electronics & Telecommunication Engineering

By

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Academic Year 2025-26

Project Approval Declaration

We,

Kshitij Anil Nanaware [Roll No:24102036]

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Pallavi Mallu Nagdhare [Roll No: 24102034]

hereby declare that we are submitting the curricular project synopsis on 'Facial Expressions Recognition System', with proper approval from our project guide & head of the department for the partial fulfillment of the requirement for the award of the degree Bachelor of Engineering in the academic year 2025-26.

Mr. P. Arotale

Guide

Department of E&TC Engineering

Dr. J. Rangole

Head

Department of E&TC Engineering

Place: Baramati

Date:

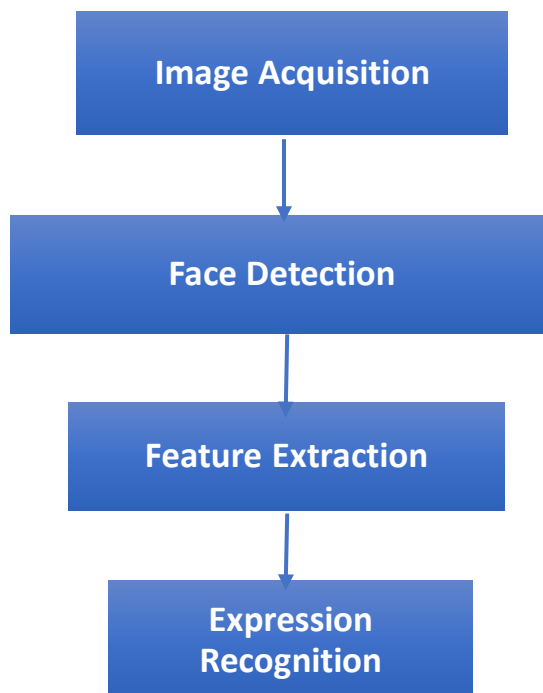
Abstract

This project presents a Facial Expressions Recognition (FER) system that uses computer vision and machine learning techniques to recognize human emotions from facial expressions. The system employs a combination of face detection, feature extraction, and classification to identify emotions such as happiness, sadness, anger, surprise, fear, and disgust. The system has potential applications in human-computer interaction, affective computing, and psychological analysis.

Introduction:

Facial expressions are a fundamental aspect of human communication, conveying emotions and intentions. Recognizing facial expressions can be a challenging task, especially in real-world scenarios with varying lighting conditions, poses, and occlusions. The Facial Expressions Recognition (FER) system is a technology that uses computer vision and machine learning techniques to recognize human emotions from facial expressions. This paper presents a FER system that uses a combination of face detection, feature extraction, and classification to identify emotions.

Block Diagram



Background of the project

- **Face Detection & Alignment:** Classical detectors (Haar cascades, HOG+SVM) gave way to deep detectors (MTCNN, RetinaFace, SSD variants) which are more robust to pose and lighting.
- **Feature Extraction & Recognition:** Early pipelines used handcrafted features (LBP, HOG) + classical classifiers. Current best practice uses CNNs (transfer learning from ImageNet or specialist FER networks) and sequence models (LSTM/Temporal CNN/CRNN) for video.
- **Datasets & Benchmarks:** Common datasets include FER2013, CK+, RAF-DB, Affect Net—used for training and evaluation and provide varied difficulty.
- **Deployment Challenges:** Generalization across subjects, handling occlusions/masks, cultural differences in expression, and real-time constraints are core issues. Temporal smoothing and multi-frame models improve stability.

Problem Statement & Proposed Project

Problem: Facial Expressions Recognition System Synopsis

Proposed Work:

- Implement a modular pipeline with interchangeable components (face detector, feature extractor, classifier).
- Baseline: LBP/HOG + SVM. Advanced: CNN-based classifier (transfer learning, optionally fine-tuned on FER datasets).
- Explore sequence modeling (temporal smoothing, LSTM/1D-CNN) to stabilize predictions in video.
- Provide GUI for live demo, recording, and CSV logging. Evaluate accuracy (per-frame and sequence), latency, and failure modes.

Significance & Objectives

Significance:

- Enables touchless emotion-aware interfaces for labs, small demonstrations, and prototype applications (e.g., driver drowsiness hinting, classroom attentiveness analytics).
- Educationally valuable: integrates CV, ML, dataset ethics, and deployment considerations.

Objectives:

- Build a working pipeline for webcam input that outputs expression labels with confidence in real time.
- Compare classical vs. deep-learning methods and report accuracy, latency (ms/frame), and resource use.
- Achieve competitive accuracy on public test sets (target: $\geq 85\%$ on FER2013-like benchmarks for selected emotion set) while maintaining responsive live performance on a mid-range laptop.
- Produce documentation, GUI, and demonstration scripts; analyze failure cases with mitigation suggestions.

Methodology

- The proposed FER system consists of the following steps:

1. Face Detection: The system uses a face detection algorithm to detect the face in an image or video.

2. Feature Extraction: The system extracts features from the detected face, such as facial landmarks, texture, and shape.

3. Classification: The system uses a machine learning algorithm, such as Support Vector Machine (SVM) or Convolutional Neural Network (CNN), to classify the extracted features into different emotions. The system is implemented using a combination of OpenCV and deep learning frameworks such as TensorFlow or PyTorch. The performance of the system is evaluated using a dataset of images or videos with labeled emotions.

Development Tools

- **Hardware:** Laptop/PC with USB webcam; optional GPU for training (Colab/GPU or local GPU).
- **Software:** Python, OpenCV, NumPy, Pandas, scikit-learn, PyTorch/TensorFlow/Keras for models; dlib/Mediapipe for landmarks; Jupyter for experiments; Streamlit/PyQt for GUI.

Proposed Project Plan

Activity \ Time	Aug 20	Oct. 20	Dec. 20	Jan. 21	Feb. 21	Mar. 21
Literature Review and Study	✓					
Mention Task 1		✓				
Mention Task 2			✓			
Mention Task 3				✓		
Mention Task 4					✓	
Final Write-up & Thesis Submission						✓

Table 1: Project Plan

References

- Goodfellow, I., Bengio, Y., & Courville, A. *Deep Learning*. MIT Press. (for DL fundamentals)
- Public FER datasets: FER2013, CK+, RAF-DB, AffectNet (for training & benchmarks).
- M. Zhang et al., “A Survey on Automatic Facial Expression Recognition,” (survey papers on FER).
- OpenCV / dlib / Mediapipe documentation for detection & landmarking. (Replace/add formal citations used during your literature review.)